

BASICS OF MONEY AND BANKING

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Money acts as a medium of exchange for economic transactions. Money was originally just metal currency, mainly gold or silver coins. The value promised by a metal currency is derived from the commodity (gold or silver) itself. It is called *commodity money*.

Nowadays, money is mainly in the form of currency notes or as legal tender issued by the Central Bank of a country. The pieces of paper on which the money is printed are not valuable in themselves. The value comes instead from the promise the government or the Central Bank makes to the person holding these notes ('Take a look at the statement from the RBI Governor on your currency note: "I promise to pay the bearer the sum of ..."). Therefore, currency notes are also called *fiat money* (money by formal authorization or decree by the government).

By the value of a currency note, we are referring to the quantities of real output that it can purchase. For instance, the value of 500 INR will be the equivalent of five decent meals (or perhaps six hair cuts, or....!) in any Indian city. The value of a currency falls during periods of inflation. That means, when prices rise, you may be able to purchase less than five meals with Rs.500. It is clear that the credibility of the Central Bank and of the government rests on their ability to maintain the value of the domestic currency (or to keep inflation under check) within certain levels.

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Financial assets or wealth is held not only in the form of coins and currency notes but also as bank deposits, bonds, insurance, pensions, options and futures. Will you like to store your wealth in the form of currency notes (money that is highly liquid), long-term fixed deposits (money that less liquid) or bonds? These decisions may vary across different population groups and over time. Some may attach importance to the 'liquidity' offered by money (it is easy to convert money into other assets or commodities) while some others may find that bonds or stocks, which are assets

offering higher ‘returns’, are more attractive. As we shall see later, such preferences will impact the effectiveness of monetary policies.

You may be interested in the history of money. Some scholars have argued that empires have been built and destroyed on successive waves of financial innovation. The emergence of Spain as a major world power coincided with that country’s control over silver mining. The Spaniards employed indigenous Indians and African slaves under extremely harsh conditions to mine silver in large quantities, especially from Peru. At the same time, however, the demise of Spain’s supremacy of the world was largely because of its failure to catch up with the new wave of financial innovation: banking. The emergence of Italy and of cities such as Amsterdam, London and Stockholm were partly due to the successes achieved by them in building banking and financial institutions

The world of money expanded with the emergence of bond markets in the 13th century and of stock markets in the 17th century. Insurance and pension funds first appeared in the 18th century; options and futures made their entry in the 19th; and a market for real estate came to be established by the 20th century (Niall Ferguson).

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The Central Bank (CB) is liable to assuring the value of each currency note it issues. Therefore, the currency notes held by the public and by the banks form the main item on the liability side of a Central Bank’s balance sheet. Against each currency note, the Central Bank holds an asset of a corresponding value. What all are assets held by a Central Bank? The Central Bank purchases bonds or debt papers issued by the government and prints currency notes having value equivalent to the bonds purchased. In fact, the Central Bank’s credit to government (or ‘printing money’) is an important form of financing the fiscal deficit (called monetization of deficit).

The Central Bank also purchases the treasury bonds or government securities issued by the governments of other countries, most importantly the US government (For a variety of economic and geopolitical reasons, the US treasury bonds or ‘dollars’ are considered as valuable as gold! – one may also ask why gold continues to be considered so valuable!). The international reserves (or foreign exchange reserves) of the Central Bank comprise treasury bonds, gold, and other foreign assets.

Central Bank Balance Sheet

Liability	Asset
1. Currency with the public (C^P)	1. Credit to government
2. Cash with commercial banks (C^B)	2. International reserves (foreign assets)
3. Commercial banks' deposits with the Central Bank	3T. Credit to commercial banks

Note: 'T' in 3T indicates the short-term or temporary nature of credit to the commercial banks.

Currency with the public (C^P) + Cash with commercial banks (C^B) = Currency in circulation

Cash with commercial banks (C^B) + Commercial banks' deposits with the Central Bank = Reserves

The sum of all items on the liability side of the Central Bank combined is referred to as the Reserve Money.

Therefore, Reserve Money or Monetary base or High powered Money or M0 is:

Currency with the public (C^P) + Cash with commercial banks (C^B) + Commercial banks' deposits with the Central Bank

or

Currency with the public + Reserves

Note that one part of the reserves is 'required reserves.' Each commercial bank is required to maintain a certain proportion of its overall (net time and demand) deposits as reserves with the Central Bank. This proportion has to be equal to or higher than the **Cash Reserve Ratio (CRR)**.

The commercial banks decide the size of the other component of the reserves (C^B) depending on the estimated cash withdrawals by their customers who hold deposits with the bank.

High powered Money or Reserve money: because the Central Bank can effect changes in Reserve money, and thereby exert control over the overall money supply.

(Overall) Money Supply (M3) =

Currency with the public + Deposits with Commercial banks.

Deposits include both demand and time deposits (that is, savings deposits and fixed deposits).

$M3 = \text{Reserve Money} * \text{Money multiplier}$

Currency with the public = Currency-Deposit Ratio (or c) * Deposits = cD

Reserves = Reserves-Deposit Ratio (or r) * Deposits = rD

$M3 = C^P + \text{Deposits} = cD + D$ Reserve Money = $C^P + \text{Reserves} = cD + rD$

Money multiplier = $(cD + D) / (cD + rD) = (c + 1) / (c + r)$

Commercial Bank Balance Sheet

Liability	Asset
1. Deposits (Demand and Time)	1. Credit to the private sector
2T.Loans from the Central Bank	2. Cash with commercial banks (C^B)
	3. Commercial banks' deposits with the Central Bank
	4. Credit to the government

Note: 'T' in 2T indicates the short-term or temporary nature of credit received by the commercial banks.

As we have seen sum of items 2 and 3 on Asset side of the above table constitutes the banks' reserves.

Item 3 is determined by CRR.

Item 4 on the asset side: is determined by SLR.

SLR Statutory liquidity Ratio: Commercial banks have been required to maintain a minimum proportion of their time and demand deposits in government securities or

other approved securities. This is called the statutory liquidity ratio. In India SLR was 20% until 1964, raised to 25% in 1964-70, 32% in 1973, 36% in 1984, and 38.5% in 1989.

In February 2022, CRR and SLR prescribed by the Reserve Bank of India are 4% and 18% respectively.

How does monetary policy work?

As we can see from the above, the Central Bank can reduce CRR or SLR, and thereby releases more funds to the commercial banks, which they can lend to the private sector.

The Central Bank can also give a short-term (“overnight”) loan to the commercial banks so that the latter can use the loan amount to increase their loans to the private sector, while maintaining their reserves with the Central Bank.

Repo rate: The rate at which the Commercial banks pay interest to the Central Bank (CB) as they borrow funds from the CB to meet their liquidity problems.

Reverse repo rate: The rate at which the Central Bank pays interest to the Commercial banks as the commercial banks park their excess funds with the CB.

‘Repo’ is repurchase agreement: The CB gives loans to the commercial banks in exchange for government securities as the collateral. These loans are for a short period (overnight) of time. At the same time, the commercial banks agree to repurchase the securities at a predetermined price and thereby pay off their loans.

‘Reverse Repo’: works in the opposite direction.

As we can see, when the Central Bank gives an overnight loan to a commercial bank, the asset side of the Central Bank and the liability side of the commercial bank will increase. As the Central bank assets increases, there will be corresponding increase in

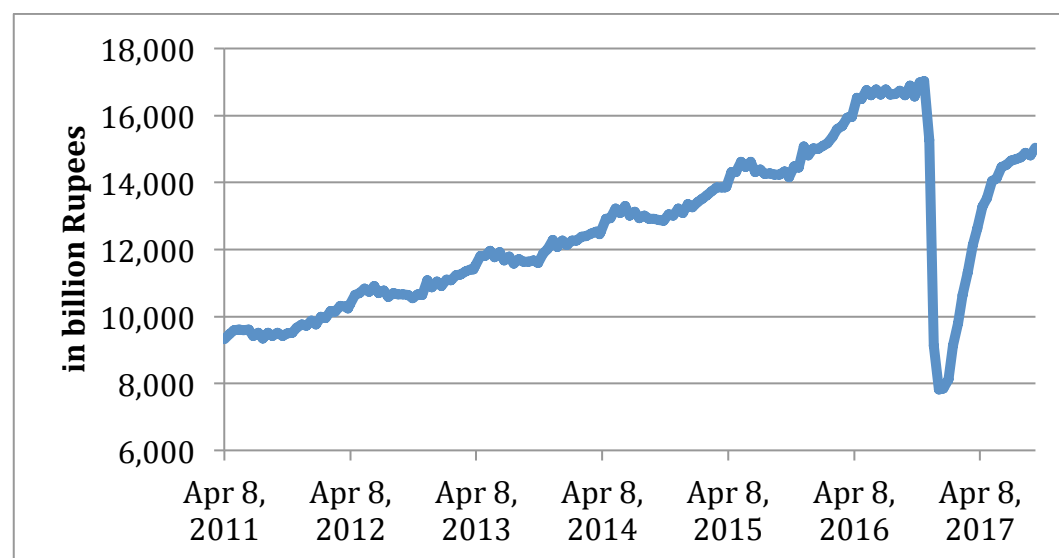
the liability side as well. Therefore, reserve money increases, and thereby overall money supply increases too.

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Another way in which the Central Bank controls reserve money is by altering the amount of deposits that commercial banks maintain with the Central Bank. When the Central Bank announces a reduction of CRR, commercial banks can reduce the share of their deposits with the Central Bank. This implies that commercial banks can then lend a greater share of their deposits as credit to the general public (households and firms). When households and firms receive money in the form of bank credit, overall money supply in the economy will go up.

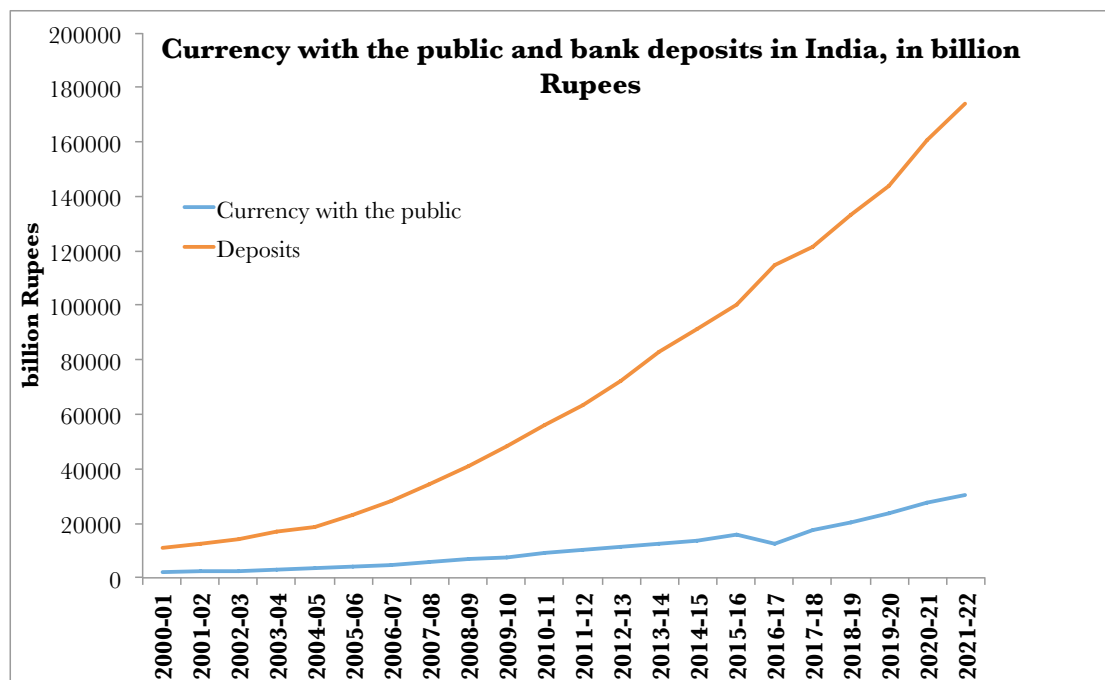
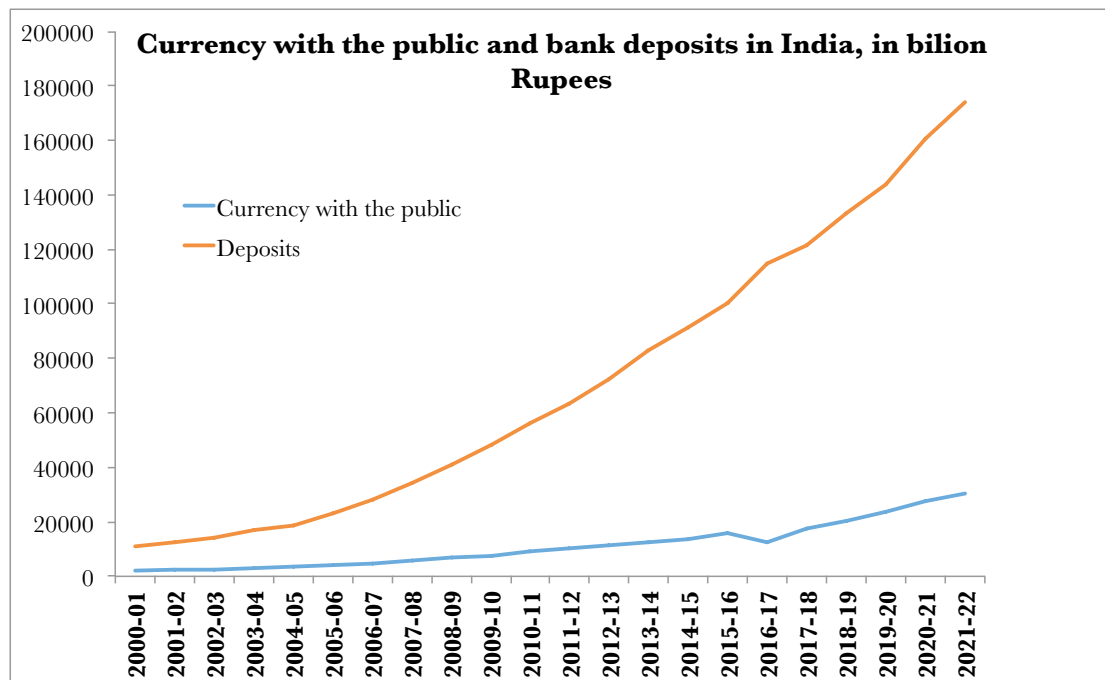
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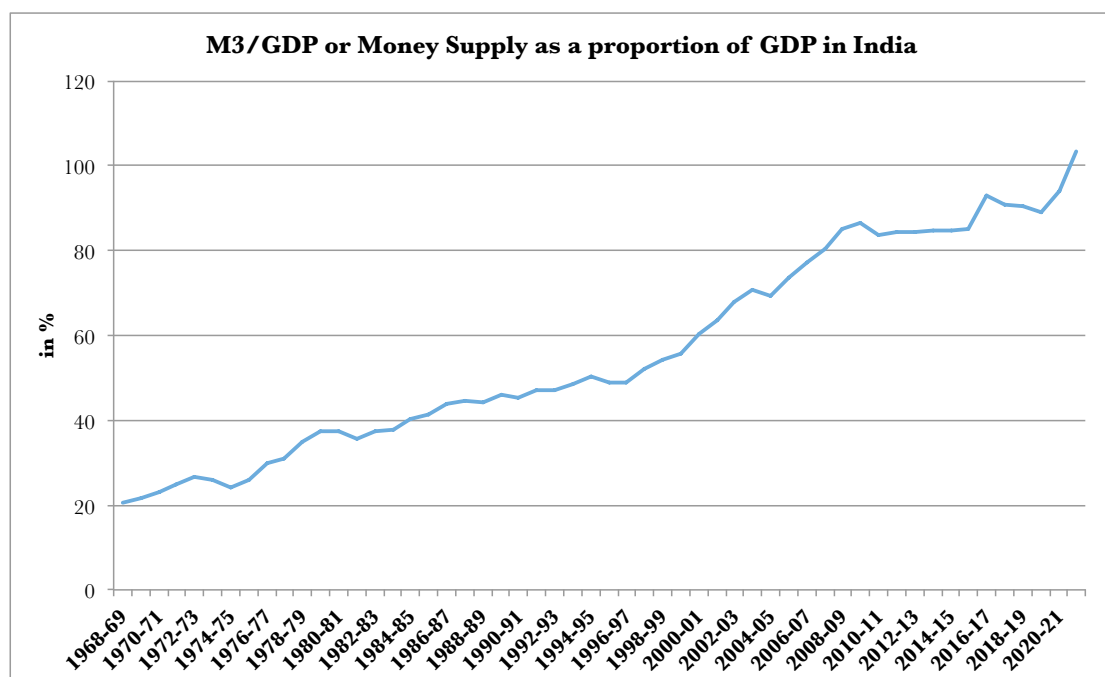
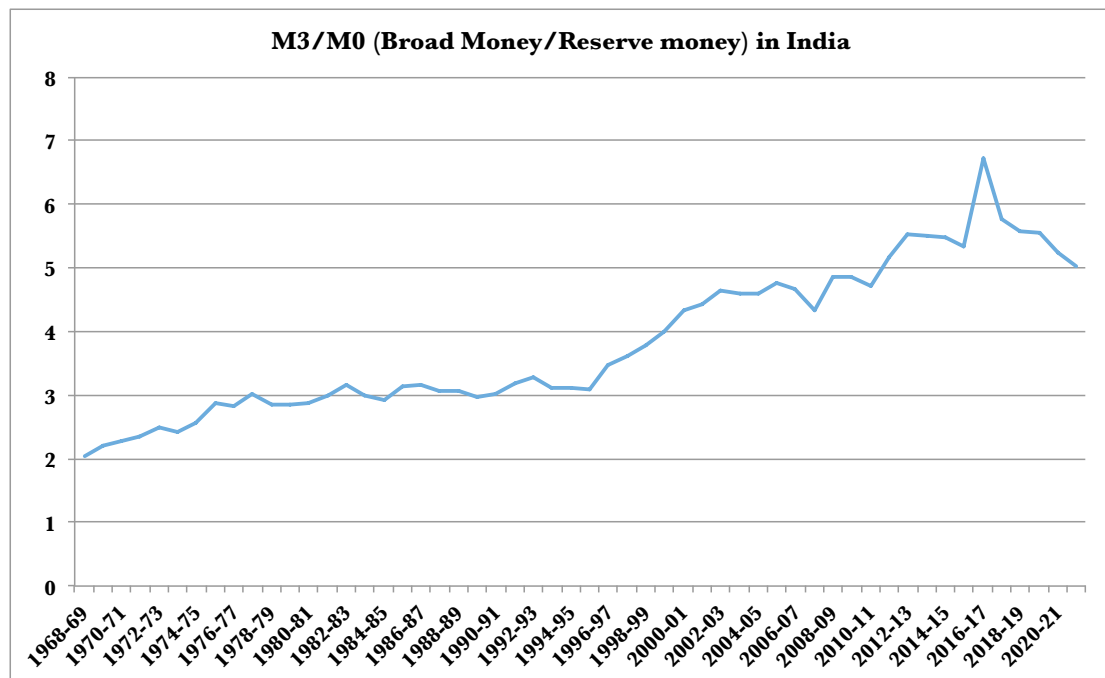
Currency with the Public in India, in billion Rupees



Source: Reserve Bank of India

See in the Figure below how currency with the public declined sharply in India after the demonetization of high value currency notes in November 2016.





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If you are interested:

How an increase in Bank credit increases the overall money supply much more:

Assume that the currency-deposit ratio (c) is equal to 0.5 and reserve-deposit ratio (r) is equal to 0.25

Then money multiplier, $(1+c)/(c+r)$, will then be equal to: $(1+0.5)/(0.5+0.25) = 2$

A money multiplier of 2 implies that when the reserve money increases by Rs. X million, total money supply in the economy increases by Rs. 2X million.

Assume that the Central Bank announces a reduction in CRR. This means that the commercial banks can reduce the amount of reserves that it holds and use that amount to provide loans to the private sector.

Assume that in the above the commercial banks can increase lending to the private sector by Rs. 150 million. Firms and individuals who receive the loans hold, say, $1/3^{\text{rd}}$ of the amount they received as currency. The rest $2/3^{\text{rd}}$ is deposited back into the banking system (assuming that the currency-deposit ratio = $1/2$)

(Consider that Mr. Y who was a recipient of the loan. He keeps a part of the amount as currency, and while figuring out what to do with the loan amount, deposited the rest with a bank. Or Mr. Y used the loan amount to make a payment to Ms. Z. Ms. Z kept a part of the receipt as currency and deposited the rest in a bank. In this way, a substantial part of the credit given by the commercial banks comes back to the banking system as deposits.

Currency-deposit Ratio is observed for the economy as whole, emerging from the preferences and actions of millions of firms and individuals.

When commercial banks lend Rs. 150 million, it may result in an increase in currency with the public by Rs. 50 million and bank deposits by Rs. 100 million.

When bank deposits increase by Rs. 100 million, Rs. 25 million will have to be kept as reserves, while the remaining Rs. 75 million could be given away to the private sector as fresh credit.

When the private sector receives Rs. 75 million as bank credit, it keeps Rs. 25 million as currency and the remaining Rs. 50 million will be back in the banking system as deposits.

And the process goes on until the commercial banks are left with no excess reserves that it can lend to the private sector.

Therefore, if the credit to private sector increases by Rs. 150 million, the overall money supply in the economy increases by Rs. 300 million.

What all happens when commercial banks have Rs.150 million of excess funds that it can lend to the private sector

Period	Deposits in excess of reserves at the beginning of the period	Credit to the private sector	Increase in currency with the public	Increase in deposits	Increase in reserves	Deposits in excess of reserves at the end of the period
1	150	150	50	100	25	75
2	75	75	25	50	12.5	37.5
3	37.5	37.5	12.5	25	6.25	18.75
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Sum for all periods		300	100	200	50	

Based on the example given in *Macroeconomics* by Errol D'Souza (2006), Pearson Education

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Other points to note:

As per the Indian definition, Reserves also include 'Other deposits with RBI')

The Central Bank also gives credit to commercial private sector. But the amounts involved are not substantial.